

Yuchen Wang

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

M.S. in Computer Science

2025 – 2027 (expected)

Champaign, IL, USA

Peking University

B.S. in Intelligent Science and Technology (Artificial Intelligence) Zhi Class (Elite Program), Top 10%

2021 – 2025

Beijing, China

INTERNSHIP EXPERIENCE

• Alibaba U.S.

Research Intern — Agentic LLM Post-Training

May 2026 – Dec 2026

Sunnyvale, CA, USA

- Core contributor to **Occamy-1.0**, a 35A3B co-work model [[Website](#) / [Paper](#)]. Architected and delivered execution-grounded data and post-training infrastructure for long-horizon tool use; the team's recipe combines SFT, HDPO, expert merging, and SAO over a **15K-trajectory, 403M-token SFT corpus**.
- Built verifier-gated pipelines for OpenClaw, Hermes, and Accio Work: token-exact replay, state recovery, and compaction-aware segments preserving tokens, log-probs, and loss masks for SFT/RL.
- Released model improves **Claw-Eval from 69.5 to 82.2 (71.4 Pass³)** and **AutomationBench from 7.5% to 27.6%**; scores 49.16 on WildClawBench and 59.0 on Terminal-Bench 2.1. Reaches the **low-cost knee of the cost-performance Pareto frontier** in the report's four-benchmark comparison.

• Freedo Technology

Research Intern — 3D Vision and Generative AI

May 2025 – Aug 2025

Beijing, China

- Built an end-to-end depth-conditioned ControlNet pipeline for 3D building reconstruction from noisy point clouds; designed depth/normal conditioning to preserve structural edges and planar surfaces, improving geometric fidelity from **32.7% to 85.3%**.

OPEN-SOURCE SYSTEMS ENGINEERING

- Numerics & training:** [FlashInfer #4984](#) fixes FP8 KV prefill calibration; contributions span [Muon numerics](#), [hybrid Mamba/MoE](#), [ms-swift packing/NCCL warmup/optimizer configuration](#), [RL logprobs](#), [GDN/Muon clipping/routing/isolation](#), [SGLang weight checks](#), and [verl FSDP validation](#).
- Performance & scale:** [vLLM #54699](#) converts FlashInfer BF16 MoE weights in-place, cutting TP2 peak allocation from **7.88 to 3.94 GiB** with earlier real 120B load/generation and 3/3 sleep-wake validation; [NeMo RL #3943](#) delivers **4.44–5.31× controlled Ray transfer speedup** while reducing driver RSS deltas to 0.1–0.3 MB; [Megatron-LM #5396/#5463](#) fuse GatedDeltaNet normalization and add Mamba recompute for long-context training.
- Agent runtime:** [Hermes #100693/#102549](#) repair schema-reference argument coercion and ownership-checked SSH backend cleanup (**22 + 14 targeted tests passed**); co-authored [NeMo Gym #2726](#) for pickle-safe cross-process failures; [SGLang #38063](#) adds cold-JIT startup diagnostics.

PAPERS & PROJECTS

• CineFlow: Dependency-Driven Parallel Video Generation [[Paper](#)]

Oct 2025 – May 2026

Advisor: Prof. Fan Lai, University of Illinois Urbana-Champaign

Tools: PyTorch, xDiT, Diffusion Transformers

- Architected CineFlow, the first dependency-driven video-diffusion inference system, after analyzing 1K+ MovieGenBench prompts (over 68% exhibit non-linear dependencies); compiled prompts into semantic DAGs, fused runtime trajectories with online edge pruning, and scheduled critical paths without retraining.
- Evaluated Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100, achieving **1.7–5.5× end-to-end speedup**, **1.30–2.02× lower P90 latency**, and **5.4–17.3% higher VBench overall**; pruning added nearly 7% throughput and improved visual coherence by 6.7–14.6%.

• Dynamic Prefill Optimization for LLM Inference [[Report](#)]

Aug 2025 – Jan 2026

Adaptive Online Packing — University of Illinois Urbana-Champaign

Tools: Python, CUDA, vLLM, Profiling

- Built an AIMD batching controller with real-time p95 TTFT feedback and greedy/DP length-aware packing before continuous batching; achieved up to **20% lower TTFT** on production-style DynamoLLM traces.

• FlashAttention-style CUDA Kernels for GPT-2 [[Report](#)]

Aug 2025 – Dec 2025

University of Illinois Urbana-Champaign

Tools: CUDA, C++, Nsight Compute/Systems

- Implemented tiled attention with online softmax and kernel fusion, raising arithmetic intensity from 15 to 157 FLOP/byte and delivering **~10× lower HBM traffic and up to ~9% end-to-end speedup**.

• RL for Legal Reasoning on Multi-Choice QA [[Thesis](#)]

Jan 2025 – Mar 2025

Advisor: Prof. Yansong Feng, Peking University

Tools: PyTorch, Qwen, GRPO

- Built a Zero-RL → distilled-CoT SFT → GRPO pipeline for structured legal reasoning, reaching **57.6% accuracy** and outperforming SFT-only and RL-only baselines.

SKILLS

Python, C++, CUDA, PyTorch, Ray, FSDP, 3D vision, point clouds, ControlNet, diffusion transformers, Megatron-LM, NeMo-RL, vLLM, SGLang, distributed training, profiling